

# LITTLE STREAM DETECTOR

## LSD 3.0 Technical Documentation

| Property                  | Value                                      |
|---------------------------|--------------------------------------------|
| Release                   | 3.0                                        |
| Edition                   | VPx completion and supported-route cleanup |
| Runtime                   | Windows PowerShell 5.1 + .NET Framework    |
| Architecture              | Single PowerShell file with embedded C#    |
| External multimedia tools | None                                       |
| Temporary media files     | None                                       |
| Documentation date        | 2 September 2026                           |

Current scope. VP8 is supported through Matroska/WebM. VP9 is supported through Matroska/WebM and standard non-fragmented MP4/M4V/MOV. VP8 routing through MP4 vp08 and VPx routing through AVI are intentionally not enabled.

Known major limit. Fragmented MP4 based on moof/traf/tfhd/tfdt/trun is not indexed in LSD 3.0.

# Contents

| Section | Topic                                   |
|---------|-----------------------------------------|
| 1       | Scope and design goals                  |
| 2       | Architecture and processing pipeline    |
| 3       | Canonical media model                   |
| 4       | Container support                       |
| 5       | Video analysis modules                  |
| 6       | VPx analysis: VP8 and VP9               |
| 7       | Audio analysis                          |
| 8       | Bitrate and quantizer analysis          |
| 9       | GUI and A/B comparison                  |
| 10      | Validation and defensive parsing        |
| 11      | Support matrix and known limitations    |
| 12      | Regression matrix and maintenance rules |
| A       | Key types and responsibilities          |
| B       | Terminology                             |

## Reading guide

Sections 1-3 describe the stable architecture. Sections 4-7 describe container and codec modules. Sections 8-10 describe measurement, presentation, and validation. Sections 11-12 define the support boundary and regression obligations.

# 1. Scope and Design Goals

Little Stream Detector 3.0 is a native Windows media inspection application. PowerShell provides orchestration and the WinForms interface, while embedded C# performs binary parsing, canonical sample construction, frame-header traversal, defensive validation, and numerical accumulation.

## 1.1 Primary objectives

- Analyze supported containers and codecs without launching FFmpeg, MediaInfo, or another multimedia executable.
- Keep container addressing separate from codec interpretation.
- Read only bounded canonical samples that have passed file-range validation.
- Report unavailable information as N/A instead of estimating or inventing values.
- Keep the user interface responsive through staged metadata preparation and background analysis.
- Support one Current A result and one immutable Reference B result for visual comparison.

## 1.2 Supported families

| Area       | Supported families                                                       |
|------------|--------------------------------------------------------------------------|
| Containers | Matroska, WebM, AVI 1.0, OpenDML AVI 2.0, MP4, M4V, MOV                  |
| Video      | H.264/AVC, H.265/HEVC, AV1, VP8, VP9, MPEG-4 Part 2/XviD                 |
| Audio      | AAC-LC, HE-AAC, HE-AACv2, MP3, PCM, AC-3, E-AC-3                         |
| Outputs    | Summary A/B, Streams, JSON, Log, bitrate profile, quantizer distribution |

## 2. Architecture and Processing Pipeline

LSD is a modular monolith. Container readers and codec analyzers are separate modules connected through a shared canonical model.

User / GUI -> native container detection -> track and sample indexing -> canonical media model -> codec analysis -> completed result snapshot -> reports and A/B graphs

| Layer                | Responsibility                                          | Representative components                              |
|----------------------|---------------------------------------------------------|--------------------------------------------------------|
| Presentation         | Window, reports, graphs, drag-and-drop, active-tab copy | PowerShell WinForms                                    |
| Coordination         | Jobs, cancellation, progress, result promotion          | Start-Meta, Start-Scan, Finish-Scan                    |
| Container parsing    | Native tracks, timing, offsets, sample sizes            | NativeMediaProbe, NativeMp4Probe, NativeAviProbe       |
| Canonical adaptation | Normalize tracks and bounded samples                    | MatroskaToLsdAdapter, Mp4ToLsdAdapter, AviToLsdAdapter |
| Codec analysis       | Configuration and frame-level syntax                    | AVC, HEVC, AV1, VP8, VP9, MPEG-4 Visual                |
| Comparison           | Freeze B and render compatible A/B series               | ReferenceResult, ReferenceOnlyView                     |

### 2.1 Processing lifecycle

| Phase | Action                           | Output                      |
|-------|----------------------------------|-----------------------------|
| 1     | Select or drop a file            | Absolute source path        |
| 2     | Detect the container             | NativeMediaInfo             |
| 3     | Parse tracks and sample indexes  | Container probe result      |
| 4     | Create bounded canonical samples | LsdMediaModel               |
| 5     | Traverse codec syntax            | Frame and audio diagnostics |
| 6     | Finalize reports and histograms  | Current A snapshot          |
| 7     | Optionally promote A to B        | Reference B snapshot        |

### 3. Canonical Media Model

The canonical layer allows the same codec parser to consume samples originating in different supported containers. A codec parser does not resolve MP4 boxes, Matroska blocks, or AVI chunks.

| Object         | Core purpose and fields                                                      |
|----------------|------------------------------------------------------------------------------|
| LsdMediaModel  | Source path, container type, file length, duration, timecode scale, tracks   |
| LsdTrackModel  | Track ID/type, codec ID/family, codec private bytes, payload format, samples |
| LsdSampleModel | File offset, length, DTS, PTS, duration, key/invisible/discardable flags     |
| LsdCodecPacket | Container-neutral packet passed to a codec-aware analyzer                    |

#### 3.1 Boundary contract

- FileOffset must be non-negative.
- Length must be positive and compatible with the in-memory parser contract.
- FileOffset + Length must remain within the source file.
- PayloadFormat must match the selected codec analyzer.

#### 3.2 Comparison snapshot

Reference B stores only completed presentation data: file identity, summary text, bitrate bins, quantizer mode, and quantizer counts. It does not keep a second active parser or source stream.

## 4. Container Support

### 4.1 Matroska and WebM

The EBML reader discovers native tracks, codec-private data, timestamps, and bounded block payloads. The VPx routes are explicit: V\_VP8 selects VP8 and V\_VP9 selects VP9.

### 4.2 Standard MP4, M4V, and MOV

The ISO Base Media reader supports non-fragmented files whose conventional sample tables are stored in moov. Supported tables include stsd, stsz, stsc, stco/co64, stts, ctts, and stss.

- VP9 sample entry vp09 is mapped to V\_VP9 and the shared VP9 parser.
- VP8 sample entry vp08 is intentionally not routed or advertised as supported.
- MOV uses the same standard ISO Base Media sample-table path as MP4/M4V.

### 4.3 AVI 1.0 and OpenDML AVI 2.0

AVI is supported for the codec combinations listed in the support matrix. VP8 and VP9 are not routed through AVI.

### 4.4 Fragmented MP4

Fragmented MP4 requires a different sample-indexing path based on moof, traf, tfhd, tfdt, and trun. LSD 3.0 does not implement that indexer.

# 5. Video Analysis Modules

| Family               | Native analysis                                                                               |
|----------------------|-----------------------------------------------------------------------------------------------|
| H.264 / AVC          | avcC and Annex B; SPS/PPS/VUI; I/P/B; SliceQP; CABAC/CAVLC                                    |
| H.265 / HEVC         | hvcC; VPS/SPS/PPS; NAL class; first-slice I/P/B; SliceQP; VUI                                 |
| AV1                  | av1C; OBU traversal; hidden/show-existing events; Base Q Index                                |
| VP8                  | Frame tag; key/inter; show_frame; boolean-coded first partition; Base Q Index                 |
| VP9                  | Superframe index; uncompressed header; reference slots; shown/hidden; KEY/INTER; Base Q Index |
| MPEG-4 Part 2 / XviD | VOL/VOP; I/P/B/S-VOP and non-coded VOPs; packed samples; VOP quantizer                        |

## 5.1 Shared output contract

Each analyzer returns frame classification, completeness diagnostics, rejected-unit information, and an optional codec-specific quantizer histogram. A histogram is shown only when the corresponding parser reports complete results.

## 5.2 Color metadata

Codec-derived color metadata is supported where the codec parser exposes it. MP4 colr/nclx and Matroska Colour are not currently used as fallback sources when codec-level signaling is absent.

## 6. VPx Analysis: VP8 and VP9

VP8 and VP9 share container-neutral routing, diagnostics, histogram calculation, GUI presentation, and A/B comparison. The bitstream parsers remain separate because the frame syntax is different.

| Feature                     | VP8                           | VP9                                          |
|-----------------------------|-------------------------------|----------------------------------------------|
| Supported containers        | Matroska/WebM                 | Matroska/WebM; standard MP4/M4V/MOV via vp09 |
| Frame prefix                | 3-byte frame tag              | Uncompressed frame header                    |
| Frame classes               | KEY / INTER                   | KEY / INTER                                  |
| Display signaling           | show_frame                    | show_frame / show_existing_frame             |
| Multi-frame sample handling | Not required by current route | VP9 superframe index                         |
| Frame-level quantizer       | Base Q Index, 0-127           | Base Q Index, 0-255                          |
| Parser core                 | VP8 boolean decoder           | Bounded MSB-first bit reader                 |

### 6.1 VP8 parser

- Validates the 3-byte frame tag and first-partition size.
- Validates key-frame start code 9D 01 2A and frame dimensions.
- Traverses the boolean-coded segmentation, loop-filter, and quantization syntax.
- Classifies KEY frames as I and INTER frames as P.
- Emits one frame-level Base Q Index from each successfully parsed coded frame.
- Reports parsed, rejected, shown, hidden, segmented, key, and inter counts.

### 6.2 VP9 parser

- Validates frame marker, profile signaling, sync code, and reference-slot requirements.
- Splits valid superframes and parses internal frames in coded order.
- Separates shown events, hidden coded frames, and show-existing events.
- Maintains reference-slot geometry for frame\_size\_from\_refs.
- Emits Base Q Index only for newly coded frames.

### 6.3 Quantizer semantics

VP8 and VP9 histograms represent a frame-level base index, not a reconstructed per-block distribution after segmentation. Higher index values generally indicate stronger quantization, but the scale is not a direct linear quality score.



## 7. Audio Analysis

| Family        | Validation and reporting                                                           |
|---------------|------------------------------------------------------------------------------------|
| AAC           | AudioSpecificConfig, core/output rate, channels, SBR, Parametric Stereo, profile   |
| MP3           | Version/layer, bitrate and sample-rate index, padding, frame size, decoded samples |
| AC-3 / E-AC-3 | Syncword, bitstream ID, sample rate, frame size, blocks, channels, LFE             |
| PCM           | Block alignment, payload divisibility, decoded duration, expected/measured bitrate |

Audio analysis is independent of the active video codec. Adding VP8 or VP9 does not introduce a second audio path and does not change Reference B semantics.

## 8. Bitrate and Quantizer Analysis

### 8.1 Bitrate profile

Every canonical video sample contributes its byte length to presentation-time bins. Hidden internal VP9 frames remain in their containing sample and do not create synthetic timing or bitrate bins.

| Codec         | Native histogram value | Unit                       |
|---------------|------------------------|----------------------------|
| AVC           | SliceQPY               | Frame e-level slice result |
| HEVC          | SliceQPY               | Frame e-level slice result |
| AV1           | Base Q Index           | New coded frame            |
| VP8           | Base Q Index           | Parsed coded frame         |
| VP9           | Base Q Index           | New internal coded frame   |
| MPEG-4 Part 2 | VOP quantizer          | Coded VOP                  |

### 8.2 Histogram display

- Current A is blue and Reference B is gold.
- Overlays require identical QpMode values.
- VP8 uses the title VP8 Base Q Index distribution and a valid domain of 0-127.
- VP9 uses the title VP9 Base Q Index distribution and a valid domain of 0-255.
- Contextual +/-1 display padding is clamped to a valid codec domain.

# 9. GUI and A/B Comparison

| View            | Content                                                       |
|-----------------|---------------------------------------------------------------|
| Summary A       | Readable current report, including codec-specific diagnostics |
| Summary B       | Frozen reference report while B exists                        |
| Streams         | Structured native track information                           |
| JSON            | Serializable current metadata and streams                     |
| Log             | Lifecycle, validation, and failure messages                   |
| Bitrate profile | Blue A and gold B series                                      |
| Quantizer panel | Mode-aware A/B statistics, bars, and percentages              |

## 9.1 State model

EMPTY -> open file -> A blue -> Set Ref B -> B gold -> open another file -> B gold + A blue -> Remove Ref B -> A blue, or EMPTY if B was the only result.

- Exactly one Current A and one Reference B are supported.
- B is immutable until removed or replaced by explicit promotion.
- B is never silently converted back into A.
- Copy follows the active report tab.

## 10. Validation and Defensive Parsing

### 10.1 General rules

- Every sample is checked against file boundaries before payload access.
- Parser readers are bounded to a sample, partition, NAL unit, OBU, or internal VP9 frame.
- Malformed syntax increments a rejected counter and preserves the first failure text.
- No guessed quantizer value is substituted after a parse failure.
- A quantizer graph is enabled only after the relevant completeness checks pass.

### 10.2 VP8 invariants

- $\text{Samples} = \text{Parsed} + \text{Rejected}$ .
- $\text{Parsed} = \text{KEY\_FRAME} + \text{INTER\_FRAME}$ .
- $\text{Shown} + \text{Hidden} = \text{Parsed}$ .
- $\text{Histogram count} = \text{Parsed}$  when  $\text{Rejected} = 0$ .

### 10.3 VP9 invariants

- $\text{InternalFrames} = \text{Parsed} + \text{ShowExisting}$ .
- $\text{Parsed} = \text{KEY\_FRAME} + \text{INTER\_FRAME}$ .
- $\text{Histogram count} = \text{Parsed}$  when  $\text{Rejected} = 0$ .
- $\text{Displayed I} + \text{P} + \text{B} + \text{Other}$  equals the container sample count for validated routes.

### 10.4 Runtime isolation

- The source is opened read-only with shared read access.
- No external multimedia process is launched.
- No temporary media file is generated during normal analysis.
- The script remains compatible with in-memory EXE hosting.

## 11. Support Matrix and Known Limitations

| Video                | Matroska/WebM | MP4/M4V/MOV | AVI          |
|----------------------|---------------|-------------|--------------|
| H.264 / AVC          | Yes           | Yes         | Yes, Annex B |
| H.265 / HEVC         | Yes           | Yes         | No           |
| AV1                  | Yes           | Yes         | No           |
| VP8                  | Yes           | No          | No           |
| VP9                  | Yes           | Yes, vp09   | No           |
| MPEG-4 Part 2 / XviD | Yes           | Yes         | Yes          |

### 11.1 Current limitations

- Fragmented MP4 is not supported. Only conventional moov-based sample tables are indexed.
- VP8 is supported only in Matroska/WebM. MP4 vp08 and AVI VP8 routes are intentionally absent.
- VP9 is supported in Matroska/WebM and standard non-fragmented MP4/MOV via vp09. VP9 in AVI is not supported.
- VP8 and VP9 histograms report a frame-level base index, not a per-block reconstruction after segmentation.
- Container color metadata is not used as fallback when codec-level color signaling is missing.
- LSD does not decode pixels and does not calculate PSNR, SSIM, VMAF, or related image metrics.
- The UI supports one A/B pair, not batch ranking.

## 12. Regression Matrix and Maintenance Rules

| Asset                               | Expected result                                                                                  |
|-------------------------------------|--------------------------------------------------------------------------------------------------|
| VP8 Matroska/WebM regression sample | 300 frames; 2 KEY; 298 INTER; 300 shown; Base Q 4-126; average 64.210000; zero rejected          |
| VP9 10-second Matroska sample       | 300 samples; 325 internal frames; 25 superframes; 2 KEY; 323 INTER; Base Q 31-179; zero rejected |
| VP9 standard MP4 sample             | 300 samples; 300 internal frames; 2 KEY; 298 INTER; Base Q 46-190; zero rejected                 |

### 12.1 Stable-code rules

- Keep VP8 and VP9 bitstream parsers independent.
- Keep Matroska/WebM V\_VP8 routing and MP4 vp09 routing explicit.
- Do not reintroduce MP4 vp08 or AVI VPx routing without a representative regression asset and an explicit support decision.
- Do not change canonical model contracts while adding a codec-specific parser.
- Preserve A/B role semantics and snapshot immutability.
- Preserve N/A reporting and rejected-unit visibility.

### 12.2 Future candidate

The remaining major container extension is fragmented MP4 sample indexing. It should be implemented as a separate indexer that resolves trex/tfhd/tfdt/trun state into the existing canonical sample model. Codec parsers should remain unchanged.

# Appendix A. Key Types and Responsibilities

| Type / state                                       | Responsibility                                            |
|----------------------------------------------------|-----------------------------------------------------------|
| NativeMediaInfo / NativeTrack                      | Container-native metadata and codec configuration         |
| LsdMediaModel / LsdTrackModel / LsdSampleModel     | Container-neutral tracks and bounded samples              |
| NativeMediaProbe / NativeMp4Probe / NativeAviProbe | Container detection, metadata, and sample indexing        |
| NativeMatroskaPacketScan                           | Shared canonical video analysis job; historical type name |
| Vp8BoolDecoder / Vp8FrameParser                    | VP8 first-partition traversal and Base Q extraction       |
| Vp9BitReader / Vp9FrameParser                      | VP9 superframe and uncompressed-header traversal          |
| CanonicalAudioAnalyzer                             | Native MP3, Dolby, PCM, and sample accounting             |
| ReferenceResult                                    | Frozen B summary, bitrate bins, QP mode, and histogram    |
| PowerShell coordinator                             | Jobs, reports, graphs, comparison, and GUI state          |

## Appendix B. Terminology

| Term               | Definition                                                                     |
|--------------------|--------------------------------------------------------------------------------|
| Canonical sample   | Container-neutral record describing a bounded payload range and timing         |
| VP8 Base Q Index   | Frame-level VP8 quantizer index in the domain 0-127                            |
| Internal VP9 frame | One coded VP9 frame extracted directly or from a superframe payload            |
| VP9 superframe     | One sample containing concatenated coded frames and a trailing size index      |
| Shown frame event  | Presentation caused by a newly coded shown frame or show_existing_frame        |
| Hidden coded frame | A newly coded frame with show_frame equal to zero                              |
| Base Q Index       | Frame-level native quantizer lookup index; 0-127 for VP8 and 0-255 for VP9/AV1 |
| Current A          | Most recently analyzed active result, shown in blue                            |
| Reference B        | Stored comparison result, shown in gold                                        |
| Complete           | All required relevant units parsed without rejection                           |

Release identification. Little Stream Detector 3.0 remains a single PowerShell 5.1 application with embedded C#, no external multimedia tools, and no temporary media files.



## Revision: Bitrate Graph Interaction

This revision documents the interactive mouse-hover tooltip added to the bitrate profile. The feature is presentation-only and uses the bitrate bins already produced by the canonical analysis pipeline.

### 8.1.1 Interactive bitrate tooltip

Moving the pointer across the bitrate profile selects one time bucket and displays the interval represented by that bucket together with the available Current A and Reference B bitrate values.

```
Time interval: 17:35 - 17:45  
A / B bitrate: 969.779 kbps | 2.344 Mbps
```

- When both series are available, values are displayed in the stable order Current A | Reference B.
- When only one series is available, the missing series is omitted rather than displayed as zero.
- The tooltip does not reopen the source, rerun a parser, or create additional bitrate data.
- The tooltip is hidden when the pointer leaves the bitrate graph.

### Bucket mapping and final-bucket handling

The horizontal pointer position is mapped to a bucket index using floor-based bucket selection. For N buckets and an active duration D, the nominal bucket duration is  $D / N$ . The final index is clamped to  $N - 1$  and the final interval ends exactly at D.

```
bucketDuration = duration / bucketCount  
start = index * bucketDuration  
end = min(duration, (index + 1) * bucketDuration)  
bitrate = bucketBytes * 8 / (end - start)
```

This model prevents the final bucket from collapsing into a zero-length interval and avoids an artificial bitrate spike caused by division by an extremely small fallback duration.

### Reference B snapshot extension

Reference B now stores media duration together with the completed summary, bitrate bins, quantizer mode, and quantizer counts. The stored duration is used only to label and calculate Reference B hover intervals correctly, including Reference-only view.

## Revision: Shared Diagnostic Helpers

A small internal cleanup consolidates deterministic statistic formatting and calculation that had been repeated across codec diagnostics. Public diagnostic properties and parser behavior remain unchanged.

### Shared helper functions

```
StatisticAverage(count, sum)
StatisticStdDev(count, sum, sumSquares)
SerializeHistogram(histogram)
```

The helpers are used by the existing AV1, VP8, VP9, MPEG-4 Visual, and SliceQPY diagnostic properties where applicable.

### Safety boundary

- Codec parsers still determine when a quantizer value is valid and when it is added.
- Histogram arrays, parsed/rejected counters, minimum and maximum values, and codec-specific validation remain independent.
- Reader position, sample boundaries, reference state, frame classification, and error recovery are not handled by the shared helpers.
- Existing PowerShell-facing property names and serialized histogram format remain unchanged.

### Updated GUI and comparison description

The bitrate profile now consists of blue Current A and gold Reference B series with an interactive time-and-bitrate tooltip. The quantizer panel remains mode-aware and continues to require compatible QpMode values for A/B overlay.

### Revision identification

| Release                 | 3.0                                                   |
|-------------------------|-------------------------------------------------------|
| Documentation revision  | 6 September 2026                                      |
| Affected area           | GUI presentation and deterministic diagnostic helpers |
| Parser changes          | None                                                  |
| Canonical model changes | None                                                  |
| External dependencies   | None                                                  |

**Release status: LSD 3.0 behavior is unchanged; this revision documents the finalized UI tooltip and safe helper c**